

Algebra & Geometry

Date: 13th Jan 2026

• **Row matrix** $[]_{1 \times n}$

• **column matrix** $[]_{n \times 1}$

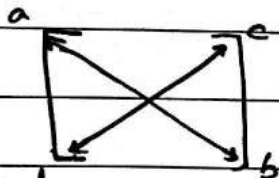
• **Null matrix** $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_3$

• **Singleton** $\begin{bmatrix} 1 \end{bmatrix}_{1,1}$

• **Square** no of rows & columns eq

• **Rectangular** no of R & C $\neq$

• **Diagonal matrix**



main diagonal $a \rightarrow b$
principle = $c \rightarrow d$

needs to be a square matrix first

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}_d$$

$\Rightarrow$ main diagonal nos should be non-zero

• **Scalar matrix**

must be a diagonal ^{matrix} first

$\Rightarrow$ main diag entries should be same.

• **Unity / Identity**

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

main diag entries should be one
 $\Rightarrow$ every identity matrix could be a diagonal matrix.

Date: _____

Symmetric Matrix

$$A^t = A$$

transpose convert R to C or vice versa

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

A

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

A^t

Skew Symmetric

$$A^t = -A$$

Triangular

upper
lower

$$\begin{bmatrix} 1 & 4 & 3 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 2 & 0 \\ 5 & 6 & 3 \end{bmatrix}$$

Conjugate

iota form change its sign

$$\begin{bmatrix} 1 & 4i & 5i \\ 0 & 2 & 4 \\ 6 & 9 & 7 \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 1 & -4i & -5i \\ 0 & 2 & 4 \\ 6 & 9 & 7 \end{bmatrix}$$

Matrix A^0

transpose of a conjugate matrix

$$\hookrightarrow (\bar{A})^t = A$$

Unitary matrix

$$A^2 = A$$

Nulpotent

$$A^2 = [0]$$

Hermitian

conjugate then transpose should be same

$$(\bar{A})^t = A$$

Quiz 6th week
 Prove if it's a matrix
 • Identity

Date:

Idempotent

$$A^2 = I$$

Periodic

$$A^{k+1}$$

Equal matrix

1 must be square, no of R & C same of both matrix

Equivalent

→ no of R & C same

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$A \neq B$$

↓
not equal but equivalent

$$1R = 2C$$

matrix matrix

the multiplication

Singular

determinant equal to zero

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

main x multiply diagonal

multiply x Principal diagonal

$$\hookrightarrow \text{Ans} = -2$$

$$1 \times 4 - 2 \times 3$$

$$4 - 6$$

$$-2$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 0 & 9 \\ 0 & 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$a(ei - fh) - b(di - fg) + c(dh - ge)$$

mighty PAPER PRODUCT

Date: _____

$$A = 1(0 \times 4 - 9 \times 3) - 2(5 \times 4 - 9 \times 0) + 3(5 \times 3 - 0 \times 0)$$

$$A = 1(0 - 27) - 2(20) + 3(15)$$

$$= -27 - 40 + 45$$

$$= -22$$

- Inverse under addition

$$2 + (-2) = 0$$

- additive inverse for every matrix possible
- multiplicative inverse not possible for every matrix

- matrix $\times$ its inverse = Identity matrix

- Inverse possible when Not Singular (non-zero determinant)

possible only

"Inverse by Row Elimination"

Date: _____

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

Operations

- $R_n \pm R_m$
 - interchange $R_n \leftrightarrow R_m$
 - $\times a \text{ no } a R_n$
 - $\div a \text{ no } \frac{1}{a} R_n$
- $a \neq 0$

$$[A \mid I]$$

$$\times R_n \times R_n$$

$$\times R_n \pm a$$

$$\times 2 \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right] \quad R_2 - 2R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right] \quad R_3 - R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right] \quad R_3 + 2R_2$$

Date: _____

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \quad -R_3$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \quad R_1 - 2R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \quad \begin{array}{l} R_1 + 9R_3 \\ R_2 + 3R_3 \end{array}$$

$$\rightarrow A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

$$A \cdot A^{-1} = I$$

Date: 20th Jan

Q //

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Solution

$$1 [(4 \times 6) - (5 \times 5)] - 2 [2 \times 6 - 5 \times 3] + 3 [2 \times 5 - 4 \times 3]$$

$$= 1(24 - 25) - 2(12 - 15) + 3(10 - 12)$$

$$= -1 + 6 - 6$$

$$= \boxed{-1} \text{ Non-singular}$$

$$[A = I]$$

$$2 \times \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 4 & 5 & 0 & 1 & 0 \\ 3 & 5 & 6 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} \\ R_2 - 2R \\ R_3 - 3R \end{array}$$

$$3 \times \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & -1 & -3 & -3 & 0 & 1 \end{array} \right] \begin{array}{l} \\ \leftarrow R_3 \quad R_2 \end{array}$$

Date: _____

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 3 & 3 & 0 & -1 \\ 0 & 0 & -1 & -2 & 1 & 0 \end{array} \right]$$

$$2 \times \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 3 & 3 & 0 & -1 \\ 0 & 0 & 1 & -2 & -1 & 0 \end{array} \right] -R_3$$

$$3 \times \left[\begin{array}{ccc|ccc} 1 & 0 & -3 & -5 & 0 & 2 \\ 0 & 1 & 3 & 3 & 0 & -1 \\ 0 & 0 & 1 & 2 & -1 & 0 \end{array} \right] R_1 - 2R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -3 & 2 \\ 0 & 1 & 0 & -3 & 3 & -1 \\ 0 & 0 & 1 & 2 & -1 & 0 \end{array} \right] \begin{array}{l} R_2 - 3R_3 \\ R_1 + 3R_3 \end{array}$$

$$A^{-1} = \left[\begin{array}{ccc} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{array} \right]$$

$$\begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Rank of Matrix

Date: 20th January

no. of non-zero rows in a matrix
convert to

upper triangular matrix

leading entry should be one

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$$

if one row of matrix is completely zero
↳ inverse nahi nikly ga

Solution =

convert to echelon form

if rank less than order to inverse nahi nikly ga

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 3 & 5 & 7 \end{bmatrix} \quad R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix} \quad R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3 - R_2$$

Rank = 2

MIGHTY PAPER PRODUCT

Date: _____

$$B = \begin{bmatrix} 2 & 2 & 4 & -8 \\ -1 & -3 & -4 & 5 \\ 2 & -1 & -1 & 0 \\ 0 & -2 & -2 & 1 \end{bmatrix} \quad \frac{1}{2} R_1$$

Solution

$$2 \times \begin{bmatrix} 1 & 1 & 2 & -4 \\ -1 & -3 & -4 & 5 \\ 2 & -1 & -1 & 0 \\ 0 & -2 & -2 & 1 \end{bmatrix} \quad R_3 \rightarrow 2R_1 - R_3$$

$$\begin{bmatrix} 1 & 1 & 2 & -4 \\ -1 & -3 & -4 & 5 \\ 0 & +1 & +1 & -8 \\ 0 & -2 & -2 & 1 \end{bmatrix} \quad R_2 + R_1$$

$$\begin{bmatrix} 1 & 1 & 2 & -4 \\ 0 & -2 & -2 & +1 \\ 0 & 1 & 1 & -8 \\ 0 & -2 & -2 & 1 \end{bmatrix} \quad R_4 \rightarrow R_2 - R_4$$

$$\begin{pmatrix} -2 & -2 \\ -2 & -2 \\ 0 & 0 \end{pmatrix} \times \begin{pmatrix} -2 \\ -2 \end{pmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 & -4 \\ 0 & -2 & -2 & +1 \\ 0 & 1 & 1 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_2 \leftrightarrow R_3$$

Date: _____

$$\times 2 \begin{bmatrix} 1 & 1 & 2 & -4 \\ 0 & 1 & 1 & -8 \\ 0 & -2 & -2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$R_3 + 2R_2$

$$\begin{bmatrix} 1 & 1 & 2 & -4 \\ 0 & 1 & 1 & -8 \\ 0 & 0 & 0 & -15 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank = 3

Answer

System of linear Algebra Eq Date: _____

Q //

$$\begin{aligned}x - y + 2z &= 3 \\x + 2y + 3z &= 5 \\3x - 4y - 5z &= -13\end{aligned}$$

Solution

$$Ax = B \quad \rightarrow \text{non-homogeneous}$$

$$\begin{bmatrix} 1 & -1 & 2 \\ 1 & 2 & 3 \\ 3 & -4 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ -13 \end{bmatrix}$$

$$[A : B]$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 1 & 2 & 3 & 5 \\ 3 & -4 & -5 & -13 \end{array} \right] \quad R_2 - R_1$$

$$\times 3 \left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 3 & 1 & 2 \\ 3 & -4 & -5 & -13 \end{array} \right] \quad R_3 - 3R_1$$

Date: _____

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 3 & 1 & 2 \\ 0 & -1 & -11 & -22 \end{array} \right]$$

$3R_3 + R_2$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & -32 & -64 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & x \\ 0 & 3 & 1 & y \\ 0 & 0 & -32 & z \end{array} \right] = \left[\begin{array}{c} 3 \\ 2 \\ -64 \end{array} \right]$$

$$x - y + 2z = 3 \rightarrow 1$$

$$3y + z = 2 \rightarrow 2$$

$$-32z = -64 \rightarrow 3$$

$$x = -1, y = 0, z = 2$$

$$2, 3y + (2) = 2 \quad 3, x - (0) + 2(2) = 3$$

$$3y = -2 + 2$$

$$x + 4 = 3$$

$$\boxed{x = -1}$$

$$y = 0/3$$

$$\boxed{y = 0}$$

Date: _____

Q.
$$\begin{aligned} 2x_1 + x_2 + 2x_3 + x_4 &= 6 \\ 6x_1 - 6x_2 + 6x_3 + 12x_4 &= 36 \\ 4x_1 + 3x_2 + 3x_3 - 3x_4 &= -1 \\ 2x_1 + 2x_2 - x_3 + x_4 &= 10 \end{aligned}$$

Sol

$$\left[\begin{array}{cccc|c} 2 & 1 & 2 & 1 & 6 \\ 6 & -6 & 6 & 12 & 36 \\ 4 & 3 & 3 & -3 & -1 \\ 2 & 2 & -1 & 1 & 10 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} & & & & \\ & & & & \\ & & & & \\ & & & & \end{array} \right]$$

Consistency of system of linear equation

↓
solution will be present

if $\text{Rank}(A) = \text{Rank}(C)$

$$[A:B] = C$$

→ Unique solution : $\text{Rank}(A) = \text{Rank}(C) = n$, no of variable

→ Infinite solution : $\text{Rank}(A) = \text{Rank}(C) < n$

→ Inconsistency : $\text{Rank}(A) \neq \text{Rank}(C)$

- Convert to arbitrary
- compare
- lower diagonal make zero
- find rank
- then categorize

$$\begin{aligned} 5x + 3y + 7z &= 4 \\ 3x + 26y + 2z &= 9 \\ 7x + 2y + 10z &= 5 \end{aligned}$$

Solution

$$Ax = B$$

$$\begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 5 \end{bmatrix}$$

sol

$$\begin{cases} 7R_2 - 3R_3 \\ 5R_3 - 7R_1 \\ 8R_3 + R_1 \end{cases}$$

Date: _____

$$[A : B] = C$$

$$\left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 3 & 26 & 2 & 9 \\ 7 & 2 & 10 & 5 \end{array} \right]$$

$$5R_2 - 3R_1$$

$$\left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 0 & 121 & -11 & 33 \\ 7 & 2 & 10 & 5 \end{array} \right]$$

$$5R_3 - 7R_1$$

$$\left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 0 & 121 & -11 & 33 \\ 0 & -11 & 1 & -3 \end{array} \right]$$

$$11R_3 + R_2$$

$$\left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 0 & 121 & -11 & 33 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\text{Rank } A = 2 \quad n = 3$$

$$\text{Rank } (C) = 2$$

Ans

$$\left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 0 & 121 & -11 & 33 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The system is consistent & infinite

MIGHTY PAPER PRODUCT

Date: _____

$$5x + 3y + 7z = 4$$

$$176y - 16z = 48$$

$$\text{let } z = k$$

$$y = \frac{48 + 16k}{176}$$

$$x = 35 \left[\frac{48 + 16k}{176} \right] + 5x + 7k = 4$$

$$x = 4 - 7k - 3 \left[\frac{48 + 16k}{176} \right] \rightarrow 4 - 7(1) - 3 \left(\frac{48 + 16}{176} \right) / 5$$

$$k = 1$$

$$z = \boxed{1}$$

$$y = \boxed{\frac{4}{11}}$$

$$x = \boxed{-\frac{9}{11}}$$

$$\rightarrow -3 - 3 \left(\frac{64}{176} \right) / 5$$

$$\rightarrow -\frac{9}{11} = -3 - 3 \left[\frac{4}{11} \right] / 5$$

$$\rightarrow \frac{-33 - 12}{11} / 5$$

$$\rightarrow \frac{-45}{11} / 5$$

$$\rightarrow -\frac{9}{11}$$

MIGHTY PAPER PRODUCT

Date: _____

$$\begin{aligned} x + 2x_2 + x_3 &= 2 \\ 3x_1 + x_2 - 2x_3 &= 1 \\ 4x - 3x_2 - x_3 &= 3 \\ 2x_1 + 4x_2 + 2x_3 &= 4 \end{aligned}$$

Solution

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & -2 \\ 4 & -3 & -1 \\ 2 & 4 & 2 \end{bmatrix} [x] = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 4 \end{bmatrix}$$

$$[A : B] = C$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 3 & 1 & -2 & 1 \\ 4 & -3 & -1 & 3 \\ 2 & 4 & 2 & 4 \end{array} \right]$$

$$2R_2 - 3R_1$$

$$R_3 - 4R_1$$

$$R_4 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & -5 & -5 & -5 \\ 0 & -11 & -5 & -35 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$3R_2$$

$$\left[\begin{array}{l} \text{Rank } A = 3 \\ \text{Rank } E = 3 \\ \eta = 3 \\ 0 \quad 0 \quad 0 \quad 0 \end{array} \right]$$

consistent and Unique

$$x + 2y + z = 2$$

$$-5y - 5z = -5$$

$$-11y - 5z = -5$$

$$y + z = 1$$

$$y = -z + 1$$

$$-11(1-z) - 5z = -5$$

$$-11 + 11z - 5z = -5$$

$$z = 1$$

$$x = 2 - 1 - 2(-z + 1)$$

$$y = -(1) + 1$$

$$y = 0$$

$$x = 2 - 2(0) = 2$$

$$x = 1$$

$$x = 1, y = 0, z = 1$$

Answer

Date: _____

Q For what value of λ and u so that the eq have

- i, no solution
- ii, unique sol
- iii) an infinite no of sol

$$2x + 3y + 5z = 9$$

$$7x + 3y - 2z = 8$$

$$2x + 3y + \lambda z = u$$

Solution = $Ax = B$

2	3	5	x	9
7	3	-2	y	8
2	3	λ	z	u

$$[A:B] = C$$

2x	2	3	5	9
	7	3	-2	8
	2	3	λ	u

$2R_2 - 7R_1$

	2	3	5	9
	0	-15	-39	-47
	2	3	λ	u

$R_3 - R_1$

Date: _____

$$\left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 0 & -15 & -39 & -47 \\ 0 & 0 & \lambda - 5 & u - 9 \end{array} \right]$$

$$A \neq C$$

$$\text{Rank } A = 2, \text{ Rank } C = 3$$

i) $\lambda = 5, u \neq 9$

no sol

ii) $\lambda \neq 5, u \neq 9$

unique sol

$$A = 3, C = 3 \Rightarrow \eta = 3$$

iii) $\lambda = 5, u = 9$

infinites

$$A = 2, 2 < \eta = 3$$

Q

$$\begin{array}{rcl} x & y & z = 6 \\ x & 2y & 3z = 10 \\ x & 2y & \lambda z = u \end{array}$$

solution

$$Ax = B$$

$$\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{array} \right] \left[\begin{array}{c} x \\ y \\ z \end{array} \right] = \left[\begin{array}{c} 6 \\ 10 \\ u \end{array} \right]$$

Date: _____

$$[A:B] = C$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & u \end{array} \right]$$

$$R_2 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 1 & 2 & \lambda & u \end{array} \right]$$

$$R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & u-6 \end{array} \right]$$

$$R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & (\lambda-1)-2 & (u-6)-4 \end{array} \right]$$

$\lambda-3 \quad u-10$

$$-\lambda + 2$$

$$-u + 10$$

i) $\lambda = 3, u \neq 10$

$$u-6$$

$$A = C \neq \eta$$

$$A = C = \eta \text{ unique}$$

ii) $\lambda \neq 3, u \neq 10$

infinite

iii) $\lambda = 3, u = 10$

MIGHTY PAPER PRODUCT

Date: _____

for what value of k the eq has a solution?

$$\begin{array}{rclcl} x & y & z & = & 1 \\ 2x & y & 4z & = & k \\ 4x & y & 10z & = & k^2 \end{array}$$

Solution

$$[A : B] = C$$

$$\times 2 \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & 1 & 4 & k \\ 4 & 1 & 10 & k^2 \end{array} \right]$$

$$R_2 - 2R_1 \rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 2 & k-2 \\ 0 & -13 & 8 & k^2-2k \end{array} \right]$$

$$R_2 - 2R_1$$

$$R_3 - 4R_2$$

$$(k^2-4) - 3k$$

$$k^2 - 3k - 4$$

$$\times 4 \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 2 & k-2 \\ 0 & 0 & 10 & k^2+k-2 \end{array} \right]$$

$$R_3 - 3R_2$$

$$4R_1 - R_3 \quad k=4, k=-1$$

$$R_3 \rightarrow R_3 - 3R_2$$

$$\text{let } k=1 \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & -8 & 4-(k^2-2k) \end{array} \right]$$

$$k^2 - 3k + 2$$

$$\leftarrow (k-1)(k-2)$$

$$k=1, k=2$$

MIGHTY PAPER PRODUCT

$$k^2 - 3k + 2$$

Date: _____

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} B = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$x + y + 2z = 1$$

$$-y = -1$$

$$2z = 0$$

$$z = 0 \rightarrow 0$$

$$y = 1$$

~~$$x + 1 + 0 = 1$$~~

$$x + 1 + 0 = 1$$

$$x = 1 - 1$$

$$x = 0$$

Answer

No solution

$$\text{Rank}(A) \neq \text{Rank } C$$

everything in inquiry except body + theory

EIGEN VALUE & EIGEN VECTORS

Date: Feb 2nd

$$Ax = B \quad \text{Augmented matrix}$$

co-efficient of efficiency
order = no of variables

$$Ax = \lambda x$$

$$Ax - \lambda x = 0$$

$$(A - \lambda I)x = 0$$

$$|A - \lambda I| = 0$$

Characteristics eq

$$I = A$$

size size

$$D = \Delta$$

$$b^2 - 4ac$$

$$\text{no of Ev} = \text{no of E}$$

$$Q \quad A = \begin{bmatrix} -4 & -6 \\ 3 & 5 \end{bmatrix}$$

Solution

$$(A - \lambda I)x = 0$$

$$|A - \lambda I| = 0$$

→ put eigenvalues here

$$\begin{bmatrix} -4 & -6 \\ 3 & 5 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

MIGHTY PAPER PRODUCT

Date: _____

$$\begin{bmatrix} -4-\lambda & -6 \\ 3 & 5-\lambda \end{bmatrix} = 0$$

$$\lambda^2 - 5\lambda - 4\lambda - 20 - 18$$

$$\lambda^2 - 9\lambda - 38 = 0$$

$$(-4-\lambda)(5-\lambda) - (3)(-6) = 0$$

$$-20 - 4\lambda + 5\lambda + \lambda^2 - 18 = 0 \quad -20 + 4\lambda - 5\lambda + \lambda^2$$

$$-20 + 4\lambda - 5\lambda + \lambda^2 + 18 = 0 \quad \lambda^2 - \lambda - 2 = 0$$

$$\lambda^2 - \lambda - 2 = 0$$

$$\lambda^2 - 2\lambda + \lambda - 2 = 0$$

$$(\lambda + 1)(\lambda - 2) = 0$$

$$[\lambda = -1, 2] \rightarrow \text{Eigen values}$$

for Eigen vectors

$$\lambda = -1$$

$$[A - \lambda I]x = 0$$

$$\begin{bmatrix} -4-\lambda & -6 \\ 3 & 5-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -3 & -6 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

Date: _____

$$-3x_1 - 6x_2 = 0 \rightarrow 1,$$

$$3x_1 + 6x_2 = 0 \rightarrow 2,$$

from 1,

$$-3x_1 = 6x_2$$

$$x_1 = -2x_2 = -2t$$

put in 2,

$$3(-2x_2) = 6x_2 = 0$$

$$-6x_2 + 6x_2 = 0$$

$$x_2 = 0$$

let $x_1 = k$

$$x_2 = -k/2$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} k \\ -k/2 \end{bmatrix} \rightarrow k \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}$$

$$\lambda = 2$$

$$\begin{bmatrix} -6 & -6 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

Date: _____

$$-6x_1 - 6x_2 = 0 \rightarrow 1$$

$$3x_1 + 3x_2 = 0 \rightarrow 2,$$

$$\text{let } x = t$$

$$3(t) + 3x_2 = 0$$

$$3t = -3x_2$$

$$x_2 = -t$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} t \\ -t \end{bmatrix} \rightarrow t \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \rightarrow \text{Eigen vector}$$

Answer

Date: _____

Q $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

Solution:

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix} - \lambda \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(3-\lambda) - (4)(2) = 0$$

$$3 - \lambda - 3\lambda + \lambda^2 - 8 = 0$$

$$3 - 4\lambda + \lambda^2 - 8 = 0$$

$$\lambda^2 - 4\lambda - 5 = 0$$

$$(\lambda - 5)(\lambda + 1)$$

$$\lambda_1 = 5, \lambda_2 = -1$$

Date: _____

for eigen vector

$$\lambda_1 = 5$$

$$\begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$-4x_1 + 4x_2 = 0 \rightarrow 1$$

$$2x_1 - 2x_2 = 0 \rightarrow 2$$

$$-4x_1 = -4x_2$$

$$x_1 = x_2$$

$$\text{let } x_1 = k$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} k \\ k \end{bmatrix} \rightarrow k \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = -1$$

Date: _____

$$\begin{bmatrix} 1 - (-1) & 4 \\ 2 & 3 - (-1) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$1 \cancel{2}x_1 + \cancel{4}x_2 = 0, \quad 2(-2x) + 4x_2 = 0$$

$$x_1 = -2x$$

$$\cancel{1} - \cancel{4}x = \cancel{4}x_2$$

$$x_2 = 1$$

$$\frac{-4}{4} = x_2$$

$$\lambda_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\lambda_1, \lambda_2 = 5, -1$$

$$x_1, x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad \underline{\text{Answer}}$$

Date: _____

Q. 8.
$$\begin{bmatrix} 2 & 0 & 1 \\ -1 & 4 & -1 \\ -1 & 2 & 0 \end{bmatrix}$$

Solution

$$\begin{bmatrix} 2 & 0 & 1 \\ -1 & 4 & -1 \\ -1 & 2 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 2-\lambda & 0 & 1 \\ -01 & 4-\lambda & -1 \\ -1 & 2 & 0-\lambda \end{bmatrix} = 0$$

$$\begin{aligned} & 2-\lambda [(4-\lambda)(0-\lambda) - (-1)(2)] - 0 [(-1)(0-\lambda) \\ & - (-1)(-1)] + 1 [(-1)(2) - (4-\lambda)(-1)] \end{aligned}$$

$$2-\lambda [-4\lambda + \lambda^2 + 2] - 0 [\lambda - 1] + 1 [-2 - (-4 + \lambda)]$$

$$-8\lambda + 4\lambda^2 + 2\lambda^2 - \lambda^3 + 4 - 2\lambda - 2 + 4 - \lambda$$

$$-\lambda^3 + 4\lambda^2 + 2\lambda^2 - 2\lambda - 8\lambda + 4 + 4 - 2 - \lambda$$

Date: _____

$$-\lambda^3 + 6\lambda^2 - 4\lambda + 6 - \lambda$$

$$-\lambda^3 + 6\lambda^2 - 3\lambda + 6 \Rightarrow \text{Synthetic}$$

~~$$-\lambda^3 + 6\lambda^2 - 3\lambda + 6$$~~

$$\bullet (2-\lambda)(\lambda^2 - 4\lambda + 2) + (2-\lambda)$$

$$\bullet (2-\lambda)(\lambda^2 - 4\lambda + 2 + 1) = 0$$

$$(2-\lambda)(\lambda^2 - 4\lambda + 3) = 0$$

$$(2-\lambda)(\lambda^2 - 3\lambda - \lambda + 3) = 0$$

$$(2-\lambda)(\lambda-3)(\lambda-1) = 0$$

$$[\lambda = 1, 2, 3] \rightarrow \text{Eigen values}$$

for eigen vectors

$$\lambda = 1$$

$$[A - \lambda I] x = 0$$

$$\begin{bmatrix} 2-1 & 0 & 1 \\ -1 & 4-1 & -1 \\ -1 & 2 & 0-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

Date: _____

$$\begin{bmatrix} 1 & 0 & 1 \\ -1 & 3 & -1 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\rightarrow \begin{bmatrix} x_1 + 0x_2 + x_3 \\ -x_1 + 3x_2 - x_3 \\ 5x_2 - x_1 - x_3 \end{bmatrix} \begin{bmatrix} -x_1 + 3x_2 - x_3 \\ -x_1 + 2x_2 - x_3 \end{bmatrix}$$

$$\rightarrow x_1 + x_3$$

$$\rightarrow -x_1 + 3x_2 - x_3$$

$$\rightarrow -x_1 + 2x_2 - x_3$$

Solved by two methods

Feb 3rd 2026

linear dependence & independence of vectors, matrices

Date:

• Vectors (matrices) $x_1, x_2, x_3 \dots x_n$ are said to be dependent if;

i) All vectors (row or column matrices) are of same order

ii) n scalar $\lambda_1, \lambda_2 \dots \lambda_n$ (not all zero) exist such that

$$\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 + \dots + \lambda_n x_n = 0$$

otherwise they are linearly independent.

Q. Examine the following vectors for linear dependent and find the relation if it exist

$$x_1 = (1, 2, 4), x_2 = (2, -1, 3), x_3 = (0, 1, 2), x_4 = (-3, 7, 2)$$

$$\lambda_1 x_1 + \lambda_2 x_2$$

Solution

$$\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 + \lambda_4 x_4 = 0$$

$$\lambda_1 (1, 2, 4) + \lambda_2 (2, -1, 3) + \lambda_3 (0, 1, 2) + \lambda_4 (-3, 7, 2) = 0$$

$$\lambda_1 + 2\lambda_2 + 0\lambda_3 - 3\lambda_4 = 0$$

$$2\lambda_1 - \lambda_2 + \lambda_3 + 7\lambda_4 = 0$$

$$4\lambda_1 + 3\lambda_2 + 2\lambda_3 + 2\lambda_4 = 0$$

MIGHTY PAPER PRODUCT

Date: _____

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 2 & -1 & 1 & 7 \\ 4 & 3 & 2 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 5 & -2 & -14 \end{bmatrix}$$

$$R_2 - 2R_1$$

$$R_3 - 4R_1$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 0 & -1 & -1 \end{bmatrix}$$

$$R_2 + R_3$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = 0$$

$$\lambda_1 + 2\lambda_2 - 3\lambda_4 = 0 \quad \rightarrow 1$$

$$-5\lambda_2 + \lambda_3 + 13\lambda_4 = 0 \quad \rightarrow 2$$

$$\lambda_3 + \lambda_4 = 0 \quad \rightarrow 3$$

from 3

$$\lambda_3 = -\lambda_4$$

$$\text{let } \lambda_3 = t,$$

$$\lambda_4 = -t$$

Date: _____

put in eq 2

$$\bullet -5\lambda_2 + t + (13t) = 0$$

$$-5\lambda_2 + t - 13t = 0$$

$$\lambda_2 = \frac{-12}{5}t$$

$$\bullet \lambda_1 = -2\left(\frac{-12}{5}t\right) + 3(-t)$$

$$\lambda_1 = \frac{9}{5}t$$

$$t = \begin{bmatrix} 9/5 \\ -12/5 \\ 1 \\ -1 \end{bmatrix}$$

$$\bullet \frac{9}{5}(1, 2, 4) - \frac{12}{5}(-2, -1, 3) + (0, 1, 2) - (-3, 7, 2) = 0$$

$$\frac{9}{5}(1) - \frac{12}{5}(2) + 0 - (-3) = 0$$

$$\frac{9}{5}(2) - \frac{12}{5}(-1) + 1 - 7 = 0$$

$$\frac{9}{5}(4) - \frac{12}{5}(3) + 2 - 2 = 0$$

dependent

Date: _____

Q. Examine for linear dependence $[1, 0, 2, 1]$, $[3, 1, 2, 1]$, $[6, 4, 2, -4]$ and $[-6, 0, -3, -4]$ and find the relation between them.

Solution

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 3 & 1 & 2 & 1 \\ 6 & 4 & 2 & -4 \\ -6 & 0 & -3 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & -2 & 2 & 6 \\ 0 & 4 & -1 & -8 \\ -6 & 0 & -3 & -4 \end{bmatrix}$$

$$2R_2 - R_3$$

$$R_3 + R_4$$

$$\begin{array}{l} \times 6 \\ \times 2 \end{array} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & -2 & 2 & 6 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 9 & 2 \end{bmatrix}$$

$$R_3 + 2R_2$$

$$R_4 + 6R_1$$

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & -2 & 2 & 6 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 0 & 10 \end{bmatrix}$$

$$R_4 - 3R_3$$

Date: _____

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & -2 & 2 & 6 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 0 & 10 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = 0$$

$$\lambda_1 + 2\lambda_3 + \lambda_4 = 0 \rightarrow i$$

$$-2\lambda_2 + 2\lambda_3 + 6\lambda_4 = 0 \rightarrow ii$$

$$3\lambda_3 + 4\lambda_4 = 0 \rightarrow iii$$

$$10\lambda_4 = 0 \rightarrow iv$$

let $\lambda_4 = t$

$$3\lambda_3 + 4t = 0$$

$$3\lambda_3 = -4t$$

$$\lambda_3 = -\frac{4}{3}t$$

$$\lambda_1 + 2\left(-\frac{4}{3}t\right) + t = 0$$

$$\lambda_1 - \frac{8}{3}t + t = 0$$

$$\lambda_1 = +\frac{8}{3}t$$

$$\lambda_1 = \frac{5}{3}t$$

$$-2\lambda_2 + 2\left(-\frac{4}{3}t\right) + 6t = 0$$

$$-2\lambda_2 + -\frac{8}{3}t + 6t = 0$$

$$-2\lambda_2 + \frac{10}{3}t = 0$$

$$\lambda_2 = -\frac{10}{3} \times \frac{1}{-2}$$

$$\lambda_2 = \frac{5}{3}t$$

MIGHTY PAPER PRODUCT

Date: _____

$$\frac{5}{3}(1, 0, 2, 1) + \frac{5}{3}(3, 1, 2, 1) - \frac{4}{3}(6, 4, 2, -4) + (-6, 0, -3, -4)$$

$$\frac{5}{3}(1) + \frac{5}{3}(3) - \frac{4}{3}(6) - 6(1) = \cancel{0} - \frac{22}{3}$$

$$\frac{5}{3}(0) + \frac{5}{3} - \frac{4}{3}(4) - 0 = \cancel{0} - \frac{11}{3}$$

$$\frac{5}{3}(2) + \frac{5}{3}(2) - \frac{4}{3}(2) - 3 = 0 - 1$$

$$\frac{5}{3}(1) + \frac{5}{3}(1) - \frac{4}{3}(-4) - 4 = \cancel{0} - \frac{24}{3}$$

linearly independent.

Cayley-Hamilton Theorem

Date: Feb 16th

$$Q. A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

solution

$$\begin{aligned} |A - \lambda I| &= 0 \\ \lambda^2 + a\lambda + b &= 0 \\ (A^2 + a\lambda + bI) &= 0 \end{aligned}$$

$$\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A \Rightarrow [A - \lambda I] = 0$$

$$A - \lambda I = 0$$

$$A^{-1} = \frac{1}{\lambda} A$$

$$\begin{bmatrix} 1-\lambda & 2 \\ 2 & -1-\lambda \end{bmatrix} = 0$$

$$A^{-1} = \frac{1}{\lambda} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

$$(1-\lambda)(-1-\lambda) - 2 \times 2 = 0$$

$$\lambda^2 - 5 = 0$$

for Cayley-Hamilton theorem

$$A^2 - 5A^{-1} = 0$$

$$A - 5A^{-1} = 0$$

$$A^{-1} = \frac{1}{5} A$$

$$A^2 - 5I = 0$$

$$I = A^{-1}$$

$$\begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A = 5A^{-1}$$

$$A^{-1} = \frac{1}{5} A$$

Hence proved $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$

$$A^2 - 5I = 0$$

Date: _____

Q2
$$\begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}$$

find the characteristic eq and also verify cayley hamilton theorem and also find A^{-1}

Solution :

$$\begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4-\lambda & 3 & 1 \\ 2 & 1-\lambda & -2 \\ 1 & 2 & 1-\lambda \end{bmatrix} = 0$$

$$\begin{aligned} & \cdot \overset{4-\lambda}{\uparrow} [(4-\lambda)(1-\lambda) - (-2)(2)] - \overset{3}{\uparrow} [(2)(1-\lambda) - (-2)(1)] + \\ & 1 [(2)(2) - (1-\lambda)(1)] \end{aligned}$$

$$\cdot [(1 - 2\lambda + \lambda^2) - (-4)] - [2 - 2\lambda - (-2)] + [4 - (1 - \lambda)]$$

$$\cdot \overset{4-\lambda}{\downarrow} [1 - 2\lambda + \lambda^2 + 4] - \overset{3}{\downarrow} [2 - 2\lambda + 2] + [4 - 1 + \lambda]$$

Date: _____

$$\bullet \quad \cancel{4+4-2-2+4-1-2\lambda+2\lambda-1+\lambda^2}$$

$$\cancel{4-\lambda+\lambda^2}$$

$$\bullet \quad (4-\lambda)[5-2\lambda+\lambda^2] - 3[4-2\lambda] + [3+\lambda]$$

$$20 - 8\lambda + 4\lambda^2 - 5\lambda + 2\lambda^2 - \lambda^3$$

$$-12 + 6\lambda$$

$$+ 3 + \lambda$$

$$\bullet \quad -\lambda^3 + 6\lambda^2 - 6\lambda + 11 = 0$$

multiply by -

$$\bullet \quad \lambda^3 - 6\lambda^2 + 6\lambda - 11 = 0$$

from Cayley Hamilton theorem

→ solving for inverse

$$A^3 - 6A^2 + 6A - 11I = 0$$

$$A(A^2 - 6A + 6I - 11A^{-1}) = 0$$

$$A^2 - 6A + 6I = 11A^{-1} = 0$$

$$11A^{-1} = A^2 - 6A + 6I$$

$$\bullet \quad A^{-1} = \frac{1}{11} (A^2 - 6A + 6I)$$

$$A^{-1} = \frac{1}{11} \left(\begin{bmatrix} 23 & 17 & -1 \\ 8 & 3 & -2 \\ 9 & 7 & -2 \end{bmatrix} - 6 \begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix} + 6 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)$$

MIGHTY PAPER PRODUCT

Date: _____

Solving to Prove

$$A^3 - 6A^2 + 6A - 11I = 0 \rightarrow \text{characteristic equation}$$

$$\begin{bmatrix} 125 & 84 & -12 \\ 36 & 23 & 0 \\ 48 & 30 & -7 \end{bmatrix} - 6 \begin{bmatrix} 23 & 17 & -1 \\ 8 & 3 & -2 \\ 9 & 7 & -2 \end{bmatrix} + 6 \begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix} - 11 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 125 & 84 & -12 \\ 36 & 23 & 0 \\ 48 & 30 & -7 \end{bmatrix} - \begin{bmatrix} +138 & +102 & 6 \\ 48 & 18 & -12 \\ 54 & 42 & -12 \end{bmatrix} + \begin{bmatrix} 24 & 18 & 6 \\ 12 & 6 & -12 \\ 6 & 12 & 6 \end{bmatrix} - \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

Hence proved

Mid Term

Problems re:
define formula &
definition

Farthest series *

HK-Dash

MIGHTY PAPER PRODUCT

"Diagonalization"

Date: April 7th 2021

$$D = P^{-1} A P$$

matrix of eigen vector

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

find eigen vectors

make matrix of eigen vectors (P)

find P^{-1}

multiply all three $P^{-1} A P$

diagonal values of new matrix would be eigen values

to

$$[A - \lambda I]$$

$$\begin{bmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{bmatrix}$$

$$\cdot (1-\lambda)[(2-\lambda)(3-\lambda) - (1)(2)] + 0 \cdot -1[(1)(2) - (2-\lambda)(2)]$$

$$\cdot (1-\lambda)[6 - 2\lambda - 3\lambda + \lambda^2 - 2] (-1)[2 - 4 - 2\lambda]$$

$$\cdot (1-\lambda)[\lambda^2 - 5\lambda + 4] + 0 \cdot -1[-2 - 2\lambda]$$

$$\cdot (1-\lambda)[\lambda^2 - 5\lambda + 4 + 2 + 2\lambda]$$

$$\cdot (1-\lambda)[\lambda^2 - 3\lambda + 6]$$

$$\cdot \lambda^2 - 3\lambda + 6 - \lambda^3 + 3\lambda^2 - 6\lambda$$

$$\cdot -\lambda^3 + 4\lambda^2 - 9\lambda + 6$$

$$\begin{bmatrix} \lambda=1 \\ \lambda=2 \\ \lambda=3 \end{bmatrix}$$

→ eigen values



MIGHTY PAPER PRODUCT



$$\frac{1}{D} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\frac{1}{D} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Date: _____

for eigen vectors

$$\begin{bmatrix} 1-1 & 0 & -1 \\ 1 & 2-1 & 1 \\ 2 & 2 & 3-1 \end{bmatrix}$$



$$\begin{bmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$-x_3 = 0$$

$$x_1 + x_2 + x_3 = 0$$

$$2x_1 + 2x_2 + 2x_3 = 0$$

$$x_1 + x_2 + 0 \rightarrow x_1 = -x_2$$

$$\text{let } x_1 = -t$$

$$-2t + 0 = -2x_2$$

$$x_2 = +1$$

$$\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1-2 & 0 & -1 \\ 1 & 2-2 & 1 \\ 2 & 2 & 3-2 \end{bmatrix}$$



$$\begin{bmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\text{let } x_3 = -t$$

$$-x_1 + 0 + (-x_3) = 0$$

$$x_1 + 0 + x_3 = 0$$

$$2x_1 + 2x_2 + x_3 = 0$$

$$\rightarrow -x_1 - x_3 = 0, x_3 = +x_1$$

$$\rightarrow -x_1 + (+t) = 0 \Rightarrow x_1 = t$$

$$\rightarrow 2t + (+t) = -2x_2$$

$$3t = -2x_2 \rightarrow x_2 = -\frac{3}{2}t$$

$$\begin{bmatrix} 1 \\ -\frac{3}{2} \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ 0 \\ +1 \end{bmatrix}$$

MIGHTY PAPER PRODUCT

Date: _____

$$\begin{bmatrix} 1-3 & 0 & -1 \\ 1 & 2-3 & 1 \\ 2 & 2 & 3-3 \end{bmatrix} \rightarrow \begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{aligned} -2x_1 - x_3 &= 0 & x_3 &= -2x_1 \\ x_1 - x_2 + x_3 &= 0 & -2x_1 + x_1 &= x_2 \Rightarrow x_2 = -x_1 \\ 2x_1 + 2x_2 &= 0 & \rightarrow x_1 &= \frac{-2x_2}{2}, \quad x_1 = -x_2 \\ & & & \text{let } x_1 = t \end{aligned}$$

$$P = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -\frac{1}{2} & -1 \\ 0 & -1 & -2 \end{bmatrix}$$

find inverse two methods $\frac{1}{D}$

$$P^{-1} = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ -1 & -\frac{1}{2} & -1 & 0 & 1 & 0 \\ 0 & -1 & -2 & 0 & 0 & 1 \end{array} \right]$$

$$P = 1(-\frac{1}{2} \times -2 - (-1)(-1)) + 1((-1)(-2) - (-1)(0)) - 1(-1 \times -1) - (-\frac{1}{2})(0)$$

$$P = 1(1 - 1) + 1(2 + 0) - 1[1 - 0]$$

$$P = 0 + 2 + (-1) \rightarrow -1$$

MIGHTY PAPER PRODUCT

$$D = P^{-1}$$

Date: _____

$$\frac{1}{b} \rightarrow \frac{1}{-1} \rightarrow -1$$

$$P^{-1} = -1 \times \begin{bmatrix} 1 & 1 & 1 \\ -1 & -\frac{1}{2} & -1 \\ 0 & -1 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & -1 & -1 \\ 1 & \frac{1}{2} & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

$$D = P^{-1} \times A \times P$$

$$\rightarrow \begin{bmatrix} -1 & -1 & -1 \\ 1 & \frac{1}{2} & 1 \\ 0 & 1 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 \\ -1 & -\frac{1}{2} & -1 \\ 0 & -1 & -2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 1 & 4 & 0 \\ 2 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

Show that $\Delta = P^{-1} A P$

Solution:

$$[A - \lambda E]$$

$$\begin{bmatrix} 1-\lambda & 0 & 0 \\ 0 & 3-\lambda & -1 \\ 0 & -1 & 3-\lambda \end{bmatrix}$$

$$1-\lambda [(3-\lambda)(3-\lambda) - (-1)(-1)] = 0 + 0$$

$$1-\lambda [9 - 3\lambda - 3\lambda + \lambda^2 - 1]$$

$$1-\lambda [8 - 6\lambda + \lambda^2]$$

$$1-\lambda [\lambda^2 - 6\lambda - 8]$$

$$(1-\lambda)(\lambda-4)(\lambda-2)$$

$$\begin{bmatrix} \lambda = 1 \\ \lambda = 2 \\ \lambda = 4 \end{bmatrix}$$

$\rightarrow$ eigen value

$$-x \frac{3}{2} + 2x^3 = 0$$

$$x^3 + 2x^3 = 0$$

$$x^3$$

Date: _____

$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 3 & -1 & -1 \\ 0 & -1 & 3 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 2 & -1 & -1 \\ 0 & -1 & 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$2x_2 - x_3 = 0 \rightarrow 2x_2 = x_3 \rightarrow x_2 = x_3/2$$

$$-x_2 + 2x_3 = 0 \rightarrow -x_3/2 + 2x_3 = 0 \rightarrow -x_3/2 + 4x_3/2 = 0 \rightarrow 3x_3/2 = 0 \rightarrow x_3 = 0$$

$$x_3 = 0$$

$$2x_2 - 0 = 0 \rightarrow x_2 = 0/2 \rightarrow 0$$

$$\begin{bmatrix} 1 & -2 & 0 & 0 \\ 0 & 3 & -2 & -1 \\ 0 & -1 & 3 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & -1 \\ 0 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$-x_1 = 0$$

$$x_2 - x_3 = 0 \rightarrow x_2 = x_3 \text{ let } x_2 = t$$

$$-x_2 + x_3 = 0 \rightarrow -(t) + x_3 = 0 \Rightarrow x_3 = t$$

$$\begin{bmatrix} 1 & -4 & 0 & 0 \\ 0 & 3 & -4 & -1 \\ 0 & -1 & 3 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} -3 & 0 & 0 & 0 \\ 0 & -1 & -1 & -1 \\ 0 & -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

-R₃

Date: _____

$$-3x_1 = 0 \rightarrow x_1 = \frac{0}{3} = 0$$

$$-x_2 - x_3 = 0$$

$$-x_2 - x_3 = 0$$

$$\rightarrow \text{let } -x_2 = x_3 \quad x_3 \rightarrow -t$$

$$-x_2 - (-t) = 0$$

$$-x_2 = -t \rightarrow x_2 = t$$

$$-t = x_3$$

$$\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$P = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & +1 & 0 & 0 & -1 \end{array} \right]$$

$-R_3$
 ~~$R_2 \rightarrow R_2 + R_3$~~

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & +1 & 0 & 1 & -1 \end{array} \right]$$

$R_3 \rightarrow R_3 + R_2$

Date: _____

$$P^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$2x - 5y = 10 \quad \dots (1)$$

$$0 = 10x -$$

$$5x = 10x - 10 \quad \dots (2)$$

$$0 = 5x - 10 \quad \dots (3)$$

$$0 = (10) - 10x \quad \dots (4)$$

$$0 = 5x - 10x \quad \dots (5)$$

$$10 = 10x - 10x \quad \dots (6)$$

$\Delta = P^{-1} A \times P$ find determinant Δ^{-1}

$$1 \begin{bmatrix} 1x-1 & -(1)1 \end{bmatrix} = 0 + 0$$

$$1 \begin{bmatrix} -1-1 \end{bmatrix} \Rightarrow -2$$

$$P^{-1} = \frac{1}{\Delta} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$P^{-1} = \frac{1}{-2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\frac{1 \times 1}{-2} = -\frac{1}{2}$$

$$\frac{-2}{-2} = 1$$

Date: _____

AP

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$AP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3+(1) & 3+(1) \\ 0 & -1+3 & -1+(-3) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 2 & -4 \end{bmatrix}$$

$P^{-1} \times AP$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 2 & -4 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & +\frac{1}{2} & +\frac{1}{2} \\ 0 & +\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

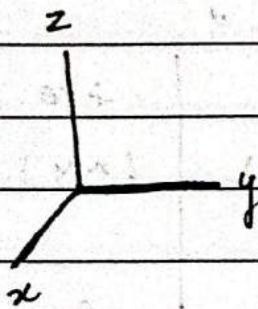
Solved

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2x-1+\frac{4x-1}{2} & \frac{2}{4}x-1+\frac{4x-1}{2} \\ 0 & 2x-1+\frac{2x-1}{2} & -\frac{2}{4}x-1+(-\frac{2}{4}) \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

2 + 2

MIGHTY PAPER PRODUCT

3D - coordinate system Date: April 27, 2026



→ volume → covered all the space $l \times w \times h$
 → area → surface level $l \times w$

• Board → 3d, can see 2 sides at the same time.

spherical

cylindrical

- radius equal from any point → ^{2D} circle, ^{3D} spherical
- radius equal from opposite sides → cylindrical



rectangular / cartesian → co-ordinates in 3D

$$(x, y, z) = (3, 4, 5)$$

→ can't be converted
 do polar bc it's 3D

polar only converted in 2D
 → (x, y)

finds polar co-ordinates
 of real numbers

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}(y/x)$$

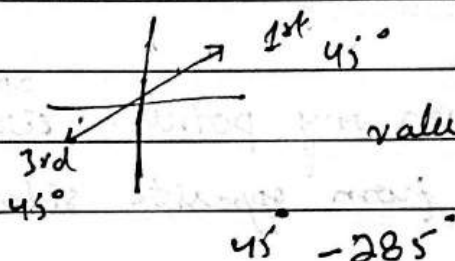
$$(x, y) = r (\cos \theta + i \sin \theta)$$

$$= (r, \theta)$$

Date: _____

1st co-ord	trig	trig	trig
2nd co-ord	cosec	sin	trig
3rd co-ord	tan	cot	trig
4th co-ord	cos	sec	trig

- sum of 2 angles 180°
- supplementary angles
- complementary angles
- sum of 2 angles $= 90^\circ$



• cylindrical formula

$$(x, y, z) = (r, \theta, z)$$

where

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}(y/x)$$

z = given co-ordinate

Date: _____

~~Q~~ $(x, y, z) = (3, 4, 5)$

Solution

$$r = \sqrt{3^2 + 4^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$r = 5$$

$$\theta = \tan^{-1} = \left(\frac{4}{3} \right) = 53^\circ$$

$$z = 5$$

$$(r, \theta, z) = (5, 53^\circ, 5) \quad \text{Answer}$$

$$\left. \begin{array}{l} x = r \cos \theta \\ y = r \sin \theta \\ z = \tan^{-1} \end{array} \right\} \text{for proof}$$

Date: _____

~~if~~ $\theta = \tan^{-1}(y/x)$

if $x > 0$ & $y > 0$ $\left[\begin{array}{c} x, y \end{array} \right]$ θ (same)

if $x < 0$ & $y > 0$ $\left[\begin{array}{c} -x, y \end{array} \right]$ $\theta - \pi$

if $x < 0$ & $y < 0$ $\left[\begin{array}{c} -x, -y \end{array} \right]$ $\pi - \theta$

if $x > 0$ & $y < 0$ $\left[\begin{array}{c} x, -y \end{array} \right]$ $-\theta$

cylindrical to cartesian

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

Date: _____

$$\underline{\underline{Q}} \quad (x, y, z) = (-3, 3, 2)$$

$$r = \sqrt{(-3)^2 + 3^2}$$

$$= \sqrt{18}$$

$$= 3\sqrt{2}$$

$$\theta = \tan^{-1} \left(\frac{3}{-3} \right) = 45^\circ \rightarrow \theta - \pi \rightarrow -135^\circ$$

$$(x, y, z) = (3\sqrt{2}, \overset{-135^\circ}{45}, 2)$$

Answer

Date: May 4th, 2026

Cartesian to spherical

$$(x, y, z) = (r, \theta, \phi)$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1}(y/x)$$

$$\phi = \cos^{-1}(z/r)$$

Q $(x, y, z) = (2\sqrt{3}, 6, -4)$

Solution

$$r = \sqrt{(2\sqrt{3})^2 + (6)^2 + (-4)^2}$$

$$\sqrt{12 + 36 + 16}$$

$$\sqrt{64}$$

$$= 8$$

$$\theta = \tan^{-1}\left(\frac{6}{2\sqrt{3}}\right)$$

$$\rightarrow \frac{3}{\sqrt{3}} \rightarrow 60^\circ \rightarrow \frac{\pi}{3}$$

$$\phi = \cos^{-1}\left(\frac{-4}{8}\right)$$

$$\rightarrow 120^\circ \rightarrow \frac{2\pi}{3}$$

$$(r, \theta, \phi) = (8, \frac{\pi}{3}, \frac{2\pi}{3})$$

Date: _____

Spherical to Cartesian

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$\Rightarrow x = \rho \sin \phi \cos \theta$$

$$8 \sin(120^\circ) \cos 60^\circ$$

$$8 \sin \frac{2\pi}{3} \cos \frac{\pi}{3}$$

$$8 \left[\frac{\sqrt{3}}{2} \times \frac{1}{2} \right]$$

$$8 \times \frac{\sqrt{3}}{2} \times \frac{1}{2} = 2\sqrt{3}$$

$$y = \rho \sin \phi \sin \theta$$

$$8 \sin 120^\circ \sin 60^\circ$$

$$\left(8 \times \frac{\sqrt{3}}{2} \right) \times \left(\frac{\sqrt{3}}{2} \right)$$

$$= 6$$

$$z = \rho \cos \phi$$

$$8 \cos 120^\circ$$

$$= -4$$

Convert

Date: May 4th, 2026

$\phi_1 (8, \pi/4, \pi/6)$ from spherical to cartesian

$\phi_2 (1, \pi/2, 1)$ from cylindrical to spherical

$\phi_3 (2, \pi/2, 1)$ from cylindrical to spherical

$\phi_4 (5, \pi/3, 2\pi/3)$ from spherical to cartesian
and cylindrical

Date: _____

$$(P, \theta, \phi) = (\cancel{54.671}, \cancel{79.91^\circ}, \cancel{56.72^\circ})$$

Answer

cylindrical to spherical $\rightarrow$ method alt

$$(r, \theta, z) \longleftrightarrow (P, \theta, \phi)$$

cyl $\rightarrow$ cartesian
cartesian $\rightarrow$ spherical

$$\theta = \theta$$

$$\phi = \tan^{-1} \left(\frac{r}{z} \right) \quad \theta/2$$

$$P = \frac{z}{\cos \phi}$$

Q $(1, \pi/2, 1)$ from cylindrical to spherical

$$(r, \theta, \phi) \Rightarrow (x, y, z)$$

$$(1, \pi/2, 1)$$

$$\phi = \frac{z}{\cos \phi} \rightarrow \frac{1}{\cos(1)}$$

Date: _____

$$\theta = 0$$

$$\phi = \frac{\pi}{2}$$

$$(r, \theta, z) = (1, \pi/2, 1)$$

$$(r, \theta, z) = (p, \theta, \phi)$$

$$r = \sqrt{x^2 + y^2}$$

$$= \sqrt{(1)^2 + (\pi/2)^2}$$

$$\sqrt{1 + 8100}$$

$$\phi = \tan^{-1}(z/r) \rightarrow \tan^{-1}\left(\frac{1}{1}\right) \rightarrow 45^\circ \rightarrow \frac{\pi}{4}$$

$$p = \frac{z}{\cos \phi} \rightarrow \frac{1}{\cos(\pi/4)} \rightarrow 1.41 \rightarrow \sqrt{2}$$

$$(r, \theta, z) = (p, \theta, \phi)$$

$$= (\sqrt{2}, \pi/2, \pi/4)$$

$$\frac{63.43 \times \pi}{180}$$

Date: _____

• $(2, \frac{\pi}{2}, 1)$ from cylindrical to spherical.

solution

$$\begin{aligned} \theta &= \phi \\ \phi &= \frac{\pi}{2} \end{aligned}$$

$$\bullet \phi = \tan^{-1}\left(\frac{z}{r}\right) \rightarrow \tan^{-1}\left(\frac{2}{1}\right) \rightarrow 63.43^\circ \rightarrow 1.107 \text{ rad}$$

$$\bullet \rho = \frac{z}{\cos \phi} \rightarrow \frac{1}{\cos(63.43)} \rightarrow 2.23 \rightarrow \frac{1}{\frac{1}{\sqrt{5}}} \rightarrow \sqrt{5}$$

$$(P, \phi, \theta) = (\sqrt{5}, 1.107, \frac{\pi}{2})$$